

# Explainability for Graph Neural Networks

## A STUDY OF PARAMETERIZED AND ALTERNATIVE APPROACHES

Base Paper: Parameterized Explainer for Graph Neural Networks  
(PGExplainer, NeurIPS 2020)

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# INTRODUCTION

## What is a GNN?

- Graphs → nodes (entities) + edges (relationships)  
Eg: users-friendships, brain regions-neural connections
- GNNs learn via message passing:
  - each node aggregates information from its neighbours
  - combines it with its own features
  - updates its representation (embedding)
  - This is repeated over several layers
- After  $L$  layers, a node's final embedding encodes information from its  $L$ -hop neighbourhood
- The learned embeddings are used for classification, regression or prediction

# INTRODUCTION

## The Black-Box Problem & What Explainability Means

- GNNs achieve strong accuracy but offer no transparency into why a prediction was made. Hence it is a black-box model
- Explainability:  
Identifying which parts of the input were responsible for a given prediction
- Two explanation questions:
  - instance-level (why this prediction, for this input?)
  - model-level (what pattern does the model show in general?)
- Relevance to our project:
  - Clinical model predicting Parkinson's stage from brain connectivity must justify which functional connections drove its decision. A black-box prediction alone isn't clinically actionable or trustworthy

# MOTIVATION

- Graph structure is discrete - standard image/text explainers (saliency, LIME) can't be reused directly
- Early explainer (GNNE explainer, Ying et al. 2019) solved this per-instance, but:
  - Retrains from scratch for every instance → expensive
  - No shared/global knowledge → poor generalization to unseen data
- **Gap:**  
Need an explainer that is efficient, inductive, and generalizes across a population

# OBJECTIVE

- Review 5 research papers spanning the GNN explainability landscape
  - model-level
  - subgraph-search
  - domain-applied survey
  - taxonomic survey
  - parameterized (base paper)
- Compare methodology, results, and limitations of each
- Justify PGExplainer as the most suitable methodology for Parkinson's Disease staging

# LITERATURE SURVEY: PAPER 1 – SubgraphX

## PAPER DETAILS:

- Name: On Explainability of Graph Neural Networks via Subgraph Explorations
- Authors: Hao Yuan, Haiyang Yu, JieWang, Kang Li, Shuiwang Ji
- Year: 2021
- Journal: Proceedings of Machine Learning Research (PMLR), Volume 139

## OBJECTIVE:

Produce a single, connected, human-intelligible explanatory subgraph instead of flat edge-importance scores

## METHODOLOGY:

- Monte Carlo Tree Search (MCTS) explores candidate subgraphs via iterative node pruning
- Each candidate scored via an efficient approximation of the Shapley value (restricted to the GNN's message-passing range for tractability)
- Best-scoring subgraph from the search tree = final explanation



# LITERATURE SURVEY: PAPER 1 - SubgraphX

## KEY RESULTS:

Consistently outperforms GNNExplainer, PGExplainer, GNN-LRP (Layer-wise Relevance Propagation adapted to GNNs) on Fidelity+/Fidelity- across Graph-SST2/5/Twitter datasets.

## ADVANTAGES:

- Best-in-class explanation quality
- Theoretically grounded (game theory)
- Clean connected-subgraph output (more human-readable than edge scores)

## LIMITATIONS:

- Computationally expensive (MCTS per instance)
- Not inductive - repeats full search for every new instance
- Impractical at scale



# LITERATURE SURVEY: PAPER 2 - XGNN

## PAPER DETAILS:

- Name: XGNN: Towards Model-Level Explanations of Graph Neural Networks
- Authors: Hao Yuan, Jiliang Tang, Xia Hu, Shuiwang Ji
- Year: 2020
- Journal: ACM International Conference on Knowledge Discovery & Data Mining

## OBJECTIVE:

Explain the model itself (not a specific input) by finding the graph pattern that maximizes a target class's predicted probability

## METHODOLOGY:

- Graph generated step-by-step via Reinforcement Learning
- Policy (GCN-based) predicts start/end node to add at each step
- Reward = classifier feedback (via rollout) + graph validity constraints (e.g: chemical valency)

Reward, piece by piece:

$$R_t = R_{t,f} \text{ (feedback reward)} + \lambda_1 \cdot (\text{rollout reward}) + \lambda_2 \cdot R_{t,r} \text{ (validity reward)}$$

- Trained via policy gradient; rolls back if reward is negative

# LITERATURE SURVEY: PAPER 2 - XGNN

## KEY RESULTS:

Recovers carbon-ring + NO<sub>2</sub>/NH<sub>2</sub> groups as mutagenic patterns on MUTAG; near-100% class probability on synthetic "Is\_Acyclic" task, using compact generated graphs

## ADVANTAGES:

- First model-level GNN explainer
- Reveals global patterns the model has learned, not just per-instance justifications

## LIMITATIONS:

- Requires hand-crafted, domain-specific validity rules
- Generated graphs may not preserve local fidelity (may not represent real inputs)
- Demonstrated only for graph classification
- RL training can be unstable

# LITERATURE SURVEY: PAPER 3 - fMRI Biomarker Systematic Review

## PAPER DETAILS:

- Name: Discovering Robust Biomarkers of Psychiatric Disorders from Resting-State Functional MRI via Graph Neural Networks: A Systematic Review
- Authors: Yi Hao Chan, Deepank Girish, Sukrit Gupta, Jing Xia, Chockalingam Kasi, Yinan He, Conghao Wang, Jagath C. Rajapakse
- Year: 2024
- Journal: NeuroImage

## OBJECTIVE:

Survey how GNN explainability is applied in practice to discover disease biomarkers (ADHD, ASD, MDD, Schizophrenia) from brain connectivity data

## METHODOLOGY:

- Reviews 65 studies under a unifying Predictor–Attributor–Evaluator (P-A-E) framework
- Predictor = trained GNN classifier
- Attributor = explainability method used
- Evaluator = robustness metrics applied



# LITERATURE SURVEY: PAPER 3 - fMRI Biomarker Systematic Review

## KEY RESULTS:

- GNNExplainer-style perturbation methods are the most-used attributor in this domain
- Only ~9% of studies (6/65) rigorously evaluated attribution robustness
- Region-level biomarkers are more reproducible than edge-level ones

## ADVANTAGES:

- Real-world, applied evidence (not just synthetic benchmarks)
- Identifies a clear research gap: model-level (XGNN) and subgraph-search (SubgraphX) methods are unused in this clinical domain

## LIMITATIONS:

- Not a method paper - no new algorithm proposed
- Highlights but doesn't resolve the lack of standardized robustness evaluation across the reviewed studies

# LITERATURE SURVEY: PAPER 4 - Explainability Taxonomic Survey

## PAPER DETAILS:

- Name: Explainability in Graph Neural Networks: A Taxonomic Survey
- Authors: Hao Yuan, Haiyang Yu, Shurui Gui, Shuiwang Ji
- Year: 2022
- Journal: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

## OBJECTIVE:

Unify all GNN explainability methods into a single taxonomy and benchmark them under common metrics

## METHODOLOGY:

- Taxonomy: instance-level (gradient-based, perturbation-based, decomposition-based) vs model-level
- Built open-source benchmark (DIG library) + new sentiment-graph datasets (Graph-SST2/5/Twitter)
- Evaluated methods on Fidelity+/-, Sparsity, Stability

# LITERATURE SURVEY: PAPER 4 - Explainability Taxonomic Survey

## KEY RESULTS:

- SubgraphX best overall Fidelity
- PGExplainer/GNNExplainer strong on node classification, weaker on graph classification vs subgraph/decomposition methods
- Older image-domain methods transfer poorly to graphs

## ADVANTAGES:

- Provides the common evidence base and metrics used across this entire seminar's comparison
- Clarifies the full landscape rather than one method

## LIMITATIONS:

- As a meta-analysis, doesn't introduce a new explainer of its own
- Benchmark conclusions are dataset-dependent and may not transfer to all domains

# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## PAPER DETAILS:

- Name: Parameterized Explainer for Graph Neural Network
- Authors: Dongsheng Luo, Wei Cheng, Dongkuan Xu, Wenchao Yu, Bo Zong, Haifeng Chen, Xiang Zhang
- Year: 2020
- Journal: Advances in Neural Information Processing Systems (NeurIPS)

## OBJECTIVE:

Learn a single, shared, parameterized function that generates instance-level explanations across an entire population - solving GNNExplainer's three core limitations:

- No global/population-level view (explains each instance independently)
- Not inductive (can't generalize to unseen nodes)
- Retrains per instance  $\rightarrow O(T \cdot |E|)$  cost



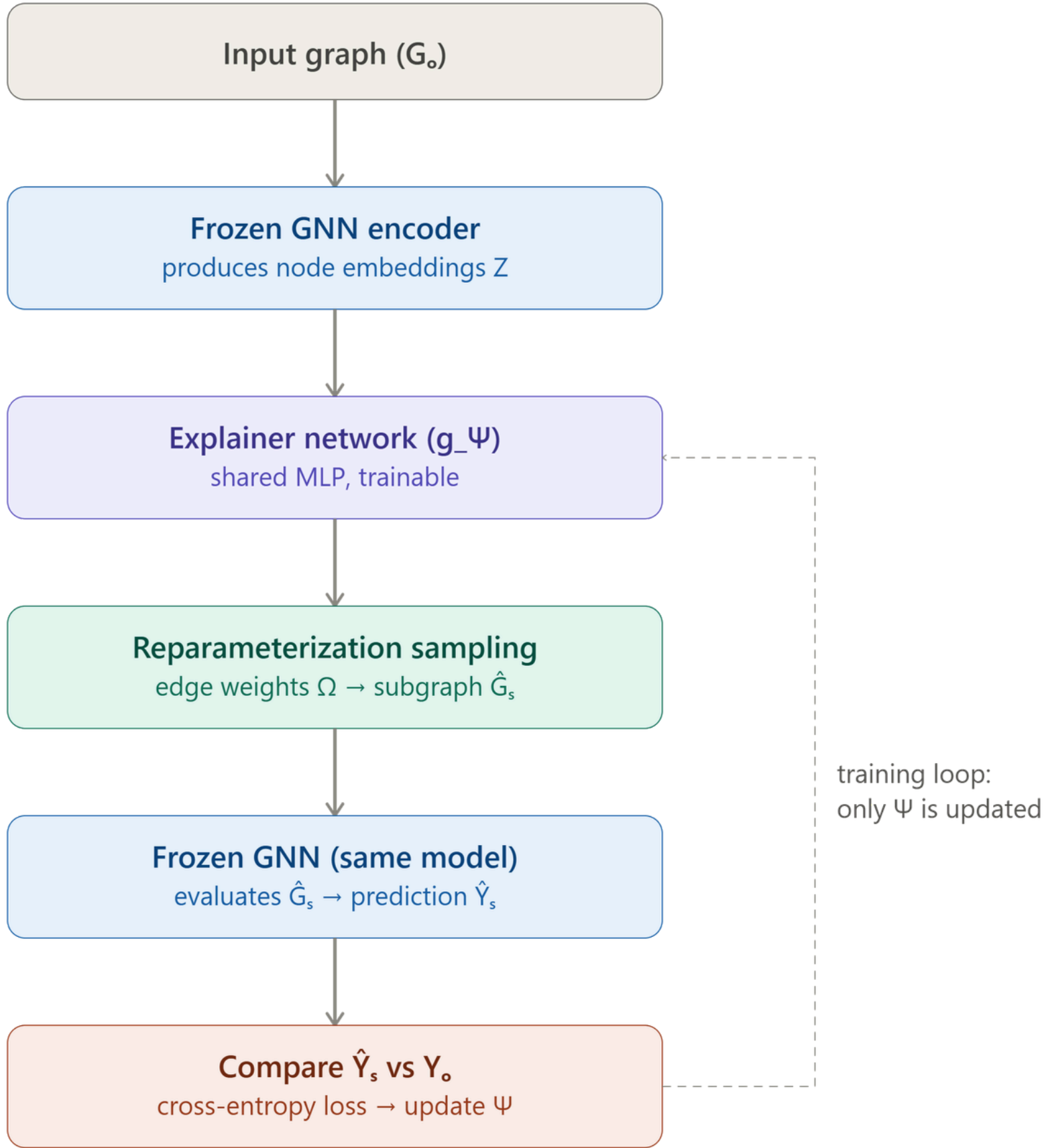
# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## KEY CONTRIBUTIONS:

- A mutual-information-based objective for identifying the explanatory subgraph
- A reparameterization trick (binary concrete distribution) making the discrete edge-selection problem differentiable
- A shared, parameterized explainer network (MLP) - trained once, generalizes to unseen instances in a single forward pass
- Demonstrated on both node classification and graph classification tasks

# LITERATURE SURVEY: PAPER 5 – Selected Paper (Base Paper)

## SYSTEM ARCHITECTURE



- frozen GNN, never updated
- trainable explainer network ( $\Psi$ )

# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## METHODOLOGY:

### Mutual Information Objective

- Split  $G_o = G_s + \Delta G$  (explanatory subgraph + remainder)
- Goal:  
Maximize  $MI(Y_o, G_s) = H(Y_o) - H(Y_o|G=G_s) \rightarrow$  reduces to minimizing conditional entropy
- Intractable over  $2^{|E|}$  subgraphs  $\rightarrow$  relax via Gilbert random graph assumption (edges as independent Bernoulli variables  $\theta_{ij}$ )

# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## METHODOLOGY:

### Reparameterization & Parameterized Network

- Binary concrete relaxation:  $\hat{e}_{ij} = \sigma((\log \varepsilon - \log(1-\varepsilon) + \omega_{ij})/\tau)$ ,  $\varepsilon \sim \text{Uniform}(0,1)$  — differentiable, discrete as  $\tau \rightarrow 0$
- $\omega_{ij}$  generated by a shared MLP:  $\Omega = g_{\Psi}(G_o, Z)$ 
  - Graph classification:  $\omega_{ij} = \text{MLP}_{\Psi}([z_i; z_j])$
  - Node classification (target node  $v$ ):  $\omega_{ij} = \text{MLP}_{\Psi}([z_i; z_j; z_v])$
- Training: minimize cross-entropy between  $Y_o$  and  $\hat{Y}_s$  (from  $K$  sampled subgraphs), averaged over the whole instance population — only  $\Psi$  is updated
- Regularizers: L1 size constraint, entropy constraint (push toward discrete 0/1), optional budget/connectivity constraints

# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## ADVANTAGES:

- Inductive (generalizes to unseen instances)
- Efficient (single forward pass)
- Population-level, more robust patterns
- Preserves local fidelity (explanation is a real subgraph of the real input, unlike XGNN)
- Works for both node and graph classification

## LIMITATIONS:

- Lower raw Fidelity than SubgraphX on some benchmarks (subgraph-search still wins on pure explanation quality)
- Explanation restricted to graph structure only (not node features)
- Assumes edge independence (Gilbert model), a simplifying approximation of true graph structure



# LITERATURE SURVEY: PAPER 5 - Selected Paper (Base Paper)

## APPLICATION & RELEVANCE TO OUR PROJECT

- Project:  
ST-GCN for Parkinson's disease staging (Healthy/Early/Mid/Late) + stability-score regression
- Every new patient's brain connectivity graph is an unseen instance, requiring an inductive explainer that generalizes to new data.
- PGExplainer learns a shared, parameterized explainer, enabling explanations for new patient scans without retraining.
- Identifies the functional brain connections that most influenced each prediction, improving interpretability, clinical trust, and decision support.

# LITERATURE SURVEY: PAPER 5 – Selected Paper (Base Paper)

## COMPARISON (Qualitative)

| Paper                       | Level                 | Explanation Target     | Core Technique                                  | Inductive? |
|-----------------------------|-----------------------|------------------------|-------------------------------------------------|------------|
| SubgraphX                   | Instance              | Connected subgraph     | MCTS + Shapley Value                            | No         |
| XGNN                        | Model                 | Generated subgraph     | RL-based generation                             | N/A        |
| fMRI Review                 | -<br>(applied survey) | -                      | P-A-E Framework                                 | -          |
| Taxonomic Survey            | -<br>(meta analysis)  | -                      | Unifying benchmark                              | -          |
| PGExplainer<br>(Base Paper) | Instance              | Edge mask/<br>subgraph | Parameterized MLP +<br>reparameterization trick | Yes        |

# LITERATURE SURVEY: PAPER 5 – Selected Paper (Base Paper)

## COMPARISON (Quantitative)

| Metric                                | GNNExplainer    | PGExplainer              | SubgraphX                | XGNN             |
|---------------------------------------|-----------------|--------------------------|--------------------------|------------------|
| AUC vs ground truth (node cls.)       | Baseline        | +4–13%                   | N/A (subgraph output)    | N/A              |
| AUC (BA-2Motifs, graph cls.)          | Lower           | +24.7% over GNNExplainer | N/A                      | N/A              |
| Inference speed (new instance)        | Slow (retrains) | Fast (12–108× faster)    | Slow (MCTS per instance) | N/A (generative) |
| Fidelity (Taxonomic Survey benchmark) | Moderate        | Moderate–High            | Highest                  | N/A              |
| Human-intelligibility                 | Medium          | Medium                   | High                     | Medium           |

# CONCLUSION

- GNN explainability spans instance-level (SubgraphX, PGExplainer) to model-level (XGNN) methods, unified under the taxonomic survey's framework
- Real-world deployment (fMRI review) shows efficiency/simplicity often matter as much as raw accuracy
- PGExplainer combines a theoretically grounded objective, a tractable training method, and a parameterized, shared architecture → inductive, efficient, generalizable
- Chosen for our project (ST-GCN, Parkinson's staging + stability regression) because every new patient scan is an unseen instance requiring an inductive explainer - a hard requirement PGExplainer satisfies directly

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